Mechanical testing of polymer composites

For understanding how these materials behave under various loads and conditions.

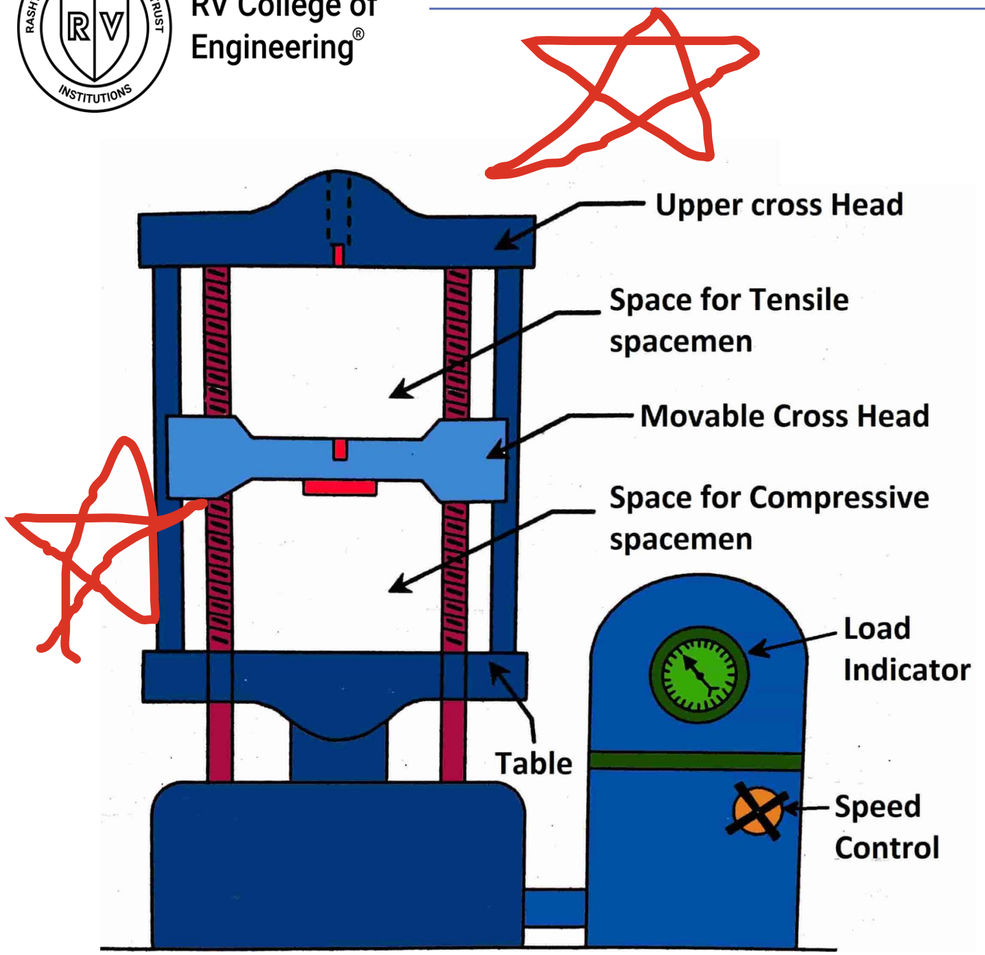
Key Mechanical Tests for Polymer Composites

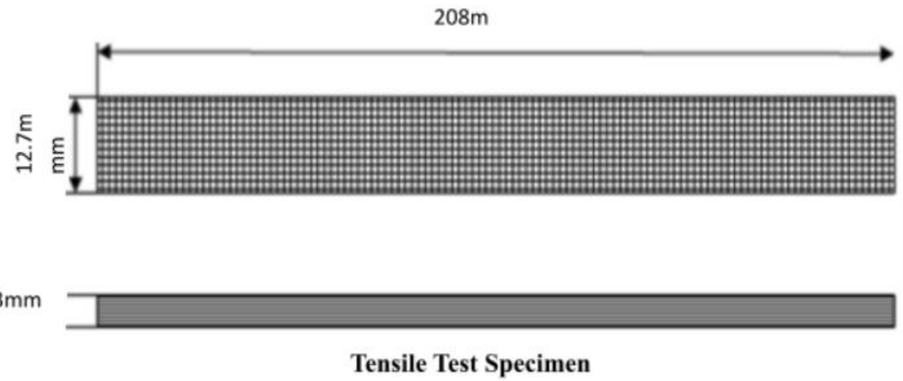
Test Type	Purpose	Common Standards
Tensile Testing	Measures strength, modulus, and elongation under uniaxial tension	ASTM D3039, ISO 527-4
Flexural Testing	Assesses bending strength and stiffness	ASTM D790, ISO 14125
Shear Testing	Determines interlaminar shear strength (ILSS)	ASTM D2344, ASTM D5379
Impact Testing	Measures energy absorption and toughness	ASTM D256, ISO 179

Why These Tests Matter

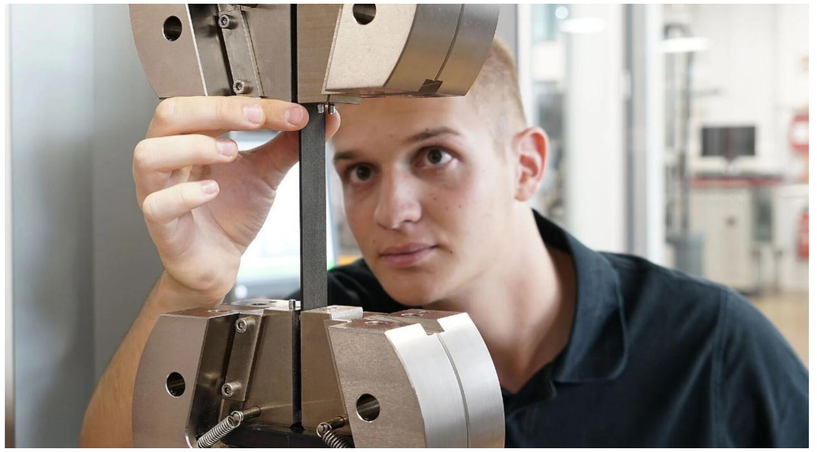
- **Material Qualification**: Ensures composites meet design specs
- **Design Validation**: Confirms structural integrity under expected loads
- **Failure Analysis**: Identifies weak points and failure modes
- **Process Optimization**: Helps refine manufacturing parameters
- **Environmental Assessment**: Evaluates performance under humidity, temperature, UV, etc.

ASTM 3039: Standard Test Method for Tensile Properties of Polymer Composite Materials





Tensile Test Specimen



ASTM D3039 tensile testing is performed using a Universal Testing Machine (UTM).

- The test specimen is positioned vertically in the grips of the testing machine.
- The grips are tightened evenly and firmly to prevent any slippage.
- The speed of testing employed is 5 mm/ min.
- Tensile testing is done with tension configuration grippers using a 10 ton capacity high precision computer controlled UTM.
- Distance between the two grippers is maintained at 50mm.
- The ultimate load carrying capacity of the composite laminates can be recorded.

The ultimate tensile strength (σ) is calculated using the formula

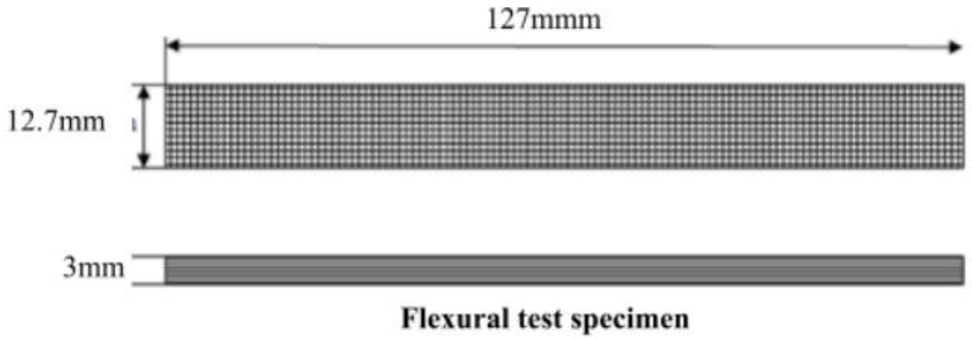
$$\sigma = P/bd \quad N/mm^2$$

Where P = maximum load in Newton's

b = width in mm

d = thickness in mm

Flexural testing is a **method used to evaluate the mechanical properties of materials subjected to bending or flexing**.



Flexural test specimen



- **The samples are cut to the dimensions per ASTM flexural testing standards.**
- **The test should be conducted at definite strain rate.**
- **Flexural test is done using a three-point bend setup.**
- **The distance between the two supports are maintained at 100mm.**
- **The ultimate load carrying capacity of the composite laminates is recorded**

The flexural strength of the composites is determined using the formula

$$\sigma_f = \frac{3PL}{2bd^2}$$

Where, P = peak load in N,

L = span length in mm,

b = width of specimen in mm,

d = thickness of specimen in mm,